

Cross-cutting Management Plan for Mangling Rangwang Chithuen NWFP Group



Gewog: Saling

Dzongkhag: Mongar

Office: Lingmithang FMU, DFO-Mongar

Plan Period: January 2025- December 2035

APPROVAL SHEET

FRPMD/FRPIS- NWFP-01/2025 (20-01-2025)

Approval Sheet

Name of NWFP Management and Marketing Group	Mangling Rangwang Chithuen Non-Wood Forest Products Management and Marketing Group
Villages(s)	Mangling
Gewog	Saling
Dzongkha/Dungkhag	Mongar
No. of households	15
Area (ha)	2472.30 ha (Cane and bamboo blocks = 319.69ha & Other NWFPs blocks = 2152.61 ha)
Plan period	January 2025 – December 2035

Prepared and submitted for Approval

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Reviewed and recommended for Approval

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Date:	6/1/25	10/1/25

APPROVED BY


Director
Department of Forests & Park Services

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PART A: NWFP MANAGEMENT AND MARKETING PLAN

1. Background

The cross-cutting management plan for sustainable management of non-wood forest products falls within the Lingmithang Forest Management Unit of Divisional Forest Office, Mongar. The Mangling village, situated in the southern part of Saling Gewog at an altitude of 1500 meters above sea level, is accessible via a 30 km farm road from Gyalpoishing town. The village consists of 15 households, with *Khengkha* as their native language. The surrounding forest is a cool broad-leaved type. However, the increasing number of cane harvests has negatively impacted the availability of cane species (*Plectocomia himalayana*, *Calamus latifolius*, and *Calamus acanthospathus*) around the village, leading to resource depletion in nearby forests. This situation highlights the need for collective responsibility to protect and sustainably use these resources.

Recognizing the economic value of NWFPs, the residents of Mangling Village have expressed interest in forming a community group to manage and trade NWFPs sustainably. This approach involves assigning management responsibility to the users themselves. Consequently, the *Mangling Rangwang Chithuen Shingmin Detshen*, a Non-Wood Forest Product Management Group, was established by 14 households. This NWFP management Unit covers an area of **6700 acres** within their traditional boundary rights, located in the Yuanri III block of the Lingmethang Forest Management Unit. The Yumbadep block covering 24.78 hectare area falls outside FMU. The group members are dedicated to the protection, conservation, sustainable management, collection, and marketing of NWFPs within their jurisdiction. They do not have rights to timber and minerals, only to NWFPs.

The specific boundaries for the collection of NWFPs are defined as follows:

-  **East:** Gangdob-Horzor-Tashithonglek
-  **West:** Laidang-Chubar-Senglengra
-  **North:** Jainala (Aentongla)-Reewalung Pong
-  **South:** Jonshingyai-Tashithongley Phu

The Mangling Rangwang Chithuen Shingmin Detshen NWFP management and marketing group was approved in 2020 and active till April 2023. In order to ensure sustainable management of cane resources the group members kept continuously vigilant and the Resungpa patrolled in the cane growing area. As the cane grows with tree support and sometimes it is difficult to pull thus, to avoid wastage the harvesting was always done in group. The group members kept continuous vigilance over the whole NWFP management area and took proper ownership of the NWFP resources. The group harvested very less resources as given in the table below as the group has not focused much on product development.

Block Name	Area (Ha)	Species	AHL (Backload)	Total Qty harvested in 3 years (BL)	Remarks
Tashithonglay	72.8	<i>Plectomia himalyana</i>	1146.60	152	Harvested for commercial purpose
		<i>Calamus acanthapothus</i>	571.20		
		<i>Neomicrocalamus andropogonifolius</i>	1021.62	15	

Its sustainable harvesting and sale of NWFPs has supplemented the income of the community members and the group members have successfully harvested and marketed several NWFPs, including cane stems,

cane shoot, cane products, mushroom, ferns, beetle leaf and damroo during the three years plan period from year 2020 to 2023. Among these, cane has been the most successful. The details of the different types of NWFPs harvested is as depicted in the table below.

Sl. No	Species harvested	Qty Harvested	Qty sold	Remarks
1	Cane stem/culm	162 Bundle	162 Bundle	
2	Cane shoot	900 Pieces	600 Pieces	
3	Yula	15 Backload	15 Backload	
4	Mushrooms	82.5 Kilograms	82.5 Kilograms	
5	Piper (pani)	80 Backload	127 Baundle	
6	<i>Diplazium</i> (ferns)	1410 Backload	1285 Bundel	
7	<i>Elatosteme</i> (damroo)	948 Backload	948 Backload	
8	Cane products			No record of the total qty and items sold. But 10% from the sale proceed Nu.18125/- is deposited in the group account.

From the income generated through sale of the above NWFPs including handicraft items made by individual members after sale 10% income **Nu.38943.5** is deducted and deposited to the group saving account. Currently the group has **Nu.52,603/=** (fifty two thousand six hundred three) only as of January, 2024 in group account maintained with Bank of Bhutan Limited. The group has not use the group fund for any expenses. The NWFP revision is aimed to maximise opportunities for NWFP market by building skills and capacities of the community members. It is also aimed to improve the economic well-being of community members ensuring the sustainability of locally available resources and the enterprise.

2. Goals and objectives:

-  To protect and conserve the NWFPs for future generations.
-  To ensure sustainable collection of locally available NWFPs through better management system
-  To generate income through the sale of NWFPs for livelihood improvement of the local people

3. Description of NWFPs:

The forest of Mangling is rich in various NWFPs, including cane species (*Plectocomia himalayana*, *Calamus acanthospathus*, *Calamus latifolius*), Yula (*Neomicrocalamus andropogonifolius*), damroo, *Rubia* spp (tsut), banana flowers (Pakpali), different types of mushrooms (Muu), *Elatostema* spp (Damru), ferns (Zeelema), bamboo species (Rawa/Glim/Daii), and *Zanthoxylum*, among others. The area is predominantly populated with *Plectocomia himalayana*, *Calamus acanthospathus*, and *Neomicrocalamus andropogonifolius*, which are especially abundant in regions like Tashithonglay, Morimodrok, Piprang, Thonglela, Ngyasergang, and Nampopong. Consequently, the group will focus on collecting cane species and Yula from the Tashithonglay area. Additionally, the group will harvest the NWFPs listed in Table 1 based on the demand from buyers and traders.

Table 1: Brief description of NWFPs for commercial harvesting

NWFP name	Harvesting area(s)(name)	Parts harvested	Use purpose (traditional & others)	Harvesting season (months)	Estimated qty. that can collected
<i>Calamus acanthospathus</i> (Krath/Minji)	Tashithonglay, Lingshibi, Morinmodrok, Khuchibrang, Senkhartoed, Ngyasrgang,	Cane	Hanicraft	November to March	As per the demand from buyers but not exceeding the AHL

	Jugbrang Nampopong, Jonshingya,				
<i>Calamus latifolius</i> (Drey/Khran)	Dakpaipho, Zangtsawa Chkphu, Ngadakla, Lenglenang, Brakgera	Cane	Hanicraft	November to March	As per the demand from buyers but not exceeding the AHL
<i>Plectocomia himalayana</i> (Rey/Phashi)	Tashithonglay (For first management plan, Morinmoddrok, Khuchibrang, Senkhartoed, Lingshibi, Ngyasrgang, Nampopong, Jugbrang, Jonshingya	Young shoot as well as matured stem	Young shoots used as vegetable. Matured stem used to produce handicraft items	Mature stems- November to March. For shoot April to August.	As per the demand from buyers but not exceeding the AHL.
Yula/Ringshoo (<i>Neomicrocalamus andropogonifolius</i>)	Dakpaipho, Changleng, Bragera, Lenglanang	Culms	Bangchungs and other handicraft items	December to April	As per the demand from buyers but not exceeding the AHL
<i>Rubia cordifolia</i> (tsod)	Dailibi, Baribang, Baylaybi, Ngadala, Khomdangbi	Entire plant	Dyes, medicine & rituals	November to February	500 kg
<i>Large bamboo sp</i>	Soinang, Dengkalibi, Khoelebrang, Kampaleng, Sengtala, Gogepong, Bogarbi, Phorzor	Culm, young shoots	Weaving, construction materials, etc. Young shoot edible	For culms: November to Feb. For shoots: June to September	As per the demand from buyers but not exceeding the AHL
Rawa (<i>Chimonobambusa callosa</i>)	Thongleya, Moringmobrok, Rawabi, Damsangla, Jainala	Culms	Flagpoles, fencing, roofing, some handicrafts	December to February	1000 Nos.
Rhue (<i>Borinda grossa</i>)	Dongpotsan, Serjachamsa,	Culms	Mats and other products	December to February	1000 Nos.
<i>Hekchung</i> (Small bamboo)	Dongpotsan,	Culms	Mats and other products	December to February	2000 Nos.
Glim (<i>Cepalstachyum</i>)	Khuchibrang, Thongleya,	Culms	Flagpoles, fencing, roofing, some handicrafts	December to February	1000 Nos.
Dai	Dalebi, Khimsar jab,	culms	Mats and other products	December to February	1000 Nos.
Damroo (<i>Elatosteme lineolatum</i>)	Zangitsawa chingku, Khalabrang, Elanchithang,	Stem and leaves	Vegetable	Throughout the year	800 bundles
Zeelema/Nakey (<i>Diplazium esculentum</i>)	Khuchibrang, Baribrang, Jukbrang	Young shoots	Vegetable	March to September	5000 bundles
Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)	Chutegang, Sangothang, Khuchibrang, Baribrang, Jubrang	Fibre	Weaving	September to October	5000 kgs
Paan (<i>Piper hamiltonii</i>)	Whole	Leaf	Stimulant	Throughout the year	1000 bundles
Thormangla (<i>Tupista nutans</i>)	Khuchibrang, Jubrang	Flower	Vegetables	October- November	500 bundles

Mushroom (Muu)	Whole	Fungi	Vegetables	April-September	1000 Kgs
Berungla (<i>Thysanolaena latifolia</i>)	Sentengla, Epaleng, Phanpong, Tokaphung	Upper parts, inflorescence	Fodder, Soft broom.	December to March	5000 bundles
Hard broom (<i>Trachycarpus fortunei</i>)	Ngadala, Tashithongley	Leaves	Broom	October-November	5000 bundles
Chaksu (Phan)	Phanpong, Epaleng, Dukulung, Tokaphung	shoot		September-October	1000 Nos.
Tsetseraba (<i>Osyris lanceolata</i>)	Whole	Leaves	Local tea leaves	May-July	500 kgs
Bamboo shoot	Soinang, Dengkalibi, Khoelebrang, Kampaleng, Sengtala, Gogepong, Bogarbi, Phorzor	Shoot	Vegetables	August-September	1500 Nos.
Chawa (<i>Zyanthozum</i>)	Thongley, Mangling area	Fruit	Essential oil, medicine, pickle & spice, water purifier	August to September	100 kgs
Walnut (Nut) (<i>Juglans regia</i>)	Khuchibrang, Jubrang	Fruit	Carnal	September to October	3000 kgs
Pilpa (<i>Piper longum</i> / <i>P. pedicelatum</i>)	Dailebi and Khuchibrang area	Fruit / spike	Medicine, sometimes used as spice	Mid-March to mid-April (<i>Piper longum</i>) Mid July to mid-September (<i>P. pedicelatum</i>)	500 kgs
Nagglabma/Yongjiba (<i>Pouzolzia sanguinea</i>)	Khuchibrang	Bark	Adhesive, Incense	August to February	1000 kgs
*Note: Not listed above can be commercialized and domestic as per the Framework					

4. Harvesting prescriptions for sustainable NWFP harvesting

4.1. Domestic purpose

Cane, bamboo (small and large), and Yula:

- Any individual harvesting NWFPs for domestic consumption must inform the Chairperson before proceeding with the collection. Harvesting cane for domestic use, particularly for the annual rimdro, should be done during the 10th month of the Bhutanese calendar.
- The maximum limit for harvesting cane shoots for domestic consumption is 10 shoots per household per season. If members collect more than this limit for marketing purposes, they will be fined according to the FNCRR 2023 regulations.
- During the times of the annual community Tshechu, in the 1st, 4th, 6th, and 8th months of the Bhutanese calendar, group members are allowed to collect 100 cane shoots, provided they inform the committee members beforehand.
- The collection of NWFPs for domestic use is free of royalty or collection fees for members. However, if found marketing these products, members will be dealt with according to the Group By-Laws and FNCRR 2023.

Others NWFPs

1. NWFPs such as mushrooms, *Elatostema* (damru), and ferns (up to 10 bundles each) for domestic consumption can be harvested at any time according to individual member needs, provided they inform the committee members before collection.

4.2. Commercial Purpose

Commercial harvesting must be conducted collectively, involving individual members from each household, and the products should be sold collectively. The group must adhere to the guidelines for harvesting and marketing NWFPs.

1. **Group Harvesting:** Commercial harvesting must be conducted collectively, involving all members.

1. **Permits for Specific NWFPs:**

- *Mushrooms, Damroo, and Ferns:* Commercial harvesting requires a collection permit from the nearby forest office.
- *Rubia cordifolia:* Harvesting must occur in the same area and at the same time, following notification from the Chairperson.
- *Cane Extraction*
 - Only members are allowed to commercially extract cane.
 - Outsiders are strictly prohibited from commercial extraction.

1. **Other NWFPs:** Members intending to harvest for commercial or domestic purposes must inform the Chairperson. The Chairperson will facilitate obtaining the necessary collection permit.

The above NWFPs should be harvested as per the harvesting prescription shown below in the table.

Table 2: Harvesting prescriptions for sustainable harvesting (As per the interim NWFPs framework, 2022)

NWFP species	Harvesting prescriptions for sustainable harvesting
Tsod (<i>Rubia cordifolia</i>)	1. Harvest stems of <i>Rubia</i> after seed dispersal, is mainly from February to April but might start as early as November in the south of Bhutan, depending on the climate and altitude 1. Use a sharp sickle or knife for the collection or harvesting of the upper part of the plant. 1. Cut the plant 5 cm above ground level. 1. Leave at least 10% of the plants without collecting or harvesting any raw materials from them (these plants shall act as seed plants for natural regeneration). 1. Do not collect roots. 1. Do not uproot entire plants.
<i>Plectocomia himalayana</i> (Ray), <i>Calamus acanthospathus</i> (Krath) and <i>Calamus latifolius</i> (Khran)	For mature stems 1. Cut cane stem only when mature, to be recognized by: 1. Loosening of leaf sheath from the lower part of the cane stem 1. Exposure of the lowest part of the stem 1. Turning to yellow reddish along the entire stem length 1. The diameter over the entire length of the cane is the same (the diameter at the lower part of the cane is reduced) 1. Pull down as much of the stem as possible (avoid leaving cane parts

	high up in the tree) ☞ Cut only canes with a usable length of at least 10 m. ☞ Discard at a maximum of 1.2 to 1.5 metres from the top end. ☞ Harvest during winter as canes have a better quality during this season (lower water content and more durable) For cane shoots ☞ Cut young shoot by bending the cane stem with the help of a long and strong stick with a hook ☞ Do not cut the entire cane for the sake of collecting shoots ☞ Cut the leading shoot with a sharp knife diagonally (not straight to avoid rotting of the remaining stump); leave lateral shoots ☞ Care shall be taken to avoid damaging sucker buds (these will later develop into lateral shoots) Guidelines for both mature stems and cane shoots ☞ Uprooting, cutting of rhizomes/suckers shall be avoided. ☞ Trees supporting the canes shall not be felled ☞ The debris after harvesting canes should be taken outside the cane area to avoid infection with insect borers or fungi
Yula/Ringshoo (<i>Neomicrocalamus andropogonifolius</i>)	☞ Harvest only culm from a clump that has more than five culms. ☞ Harvest culm when it produces a secondary shoot at the top. ☞ Retain at least two to three young healthy culms in each clump to favour more shooting in the following season.
Rawa (<i>Chimonobambusa callosa</i>); Rhue (<i>Borinda grossa</i>) & Glim	☞ Cut the culm above the first internode from the base. ☞ Leave bamboo shoot or culm of less than one year old intact. ☞ Cut at a maximum of 33% of the 1+ year and 2+ year old culm per clump. ☞ Do not cut culm when new shoots are sprouting.
Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)	☞ Peel out fibre after cutting the plant from the base. ☞ Harvest plant only after seed dispersal. ☞ Do not collect all plants from one area.
Zeelema/Nakey (<i>Diplazium esculentum</i>)	☞ Harvests young shoots that are 15cm in length. ☞ Don't harvest all shoots coming up in season. ☞ Retain at least 50% of the the young shoot of the last flush for the development of leaves. ☞ Don't harvest the shoots that have developed into leaves.
Walnut (Nut) (<i>Juglans regia</i>)	☞ Hand-pick fruits only when fallen on the ground, or ☞ Use long bamboo poles to shake off fruits from the trees. ☞ Don't cut tree or branches to collect fruits.
Damroo (<i>Elatosteme lineolatum</i>)	☞ Collect plants that are 15cm and above. ☞ Retain stump of around 3" for sprouting of new shoots. ☞ Don't uproot the whole plant. ☞ Shouldn't cut down trees and create more openings. ☞ Shouldn't disturb its catchment area.

Nagglabma/Yongjiba (<i>Pouzolzia sanguinea</i>)	☞ Cut the plant 10 cm above the ground level. ☞ Do not damage the remaining young stands/ seedlings/ saplings. ☞ Should not cut down surrounding trees to ease harvesting. ☞ Leave at least 10% of the plants to allow natural regeneration. ☞ Harvesting of roots is strictly prohibited for ecological reasons
Paan (<i>Piper hamiltonii</i>)	☞ Collect leaves only by hand picking without damaging the climber and its tender buds. ☞ Harvesting of leaves from a single (same) climber shall be allowed to a maximum of 4 times per year. ☞ Collect a maximum of 80% of the leaves from a single climber. ☞ Do not cut down the host tree to ease harvesting.
Thormangla (<i>Tupista nutans</i>)	☞ Collect only flowers by hand. ☞ Do not uproot mother plants. ☞ Do not expose plant root to direct sunlight.
Raktang/Pilpa (<i>Piper mullesea</i>)	☞ Collect fruits only when fully mature. ☞ Harvest fruits only after the announced starting date for harvesting (The date is usually announced by the coordinator in September). ☞ Harvest fruits only by hand picking without damaging, cutting or uprooting the entire plant
Chawa (<i>Zyanthozum</i>)	☞ Harvest fruit by handpicking, if possible,climbing on trees. ☞ Use long bamboo poles with a hook to facilitate picking of the fruit. ☞ Collect only 70% of the fruits and leave 30% to allow natural regeneration. ☞ Do not cut down the tree for collection of fruit.
Berungla (<i>Thysanolaena latifolia</i>)	☞ Do not damage the rhizomes in order to allow natural regeneration ☞ Cut the matured stem right from base
Hard broom (<i>Trachycarpus fortunei</i>)	☞ Do not cut the whole plant. ☞ Cut matured branches and leave seeds for regeneration. ☞ Retain immature branches.
Tsetseraba (<i>Osyris lanceolata</i>)	☞ Collect only leaves by lopping outgrown branches. ☞ Do not cut down trees.
Dhai	☞
Homestead bamboos (<i>Bambusa nutans</i> , <i>Dendrocalamus hamiltonii</i> & <i>D. sikkimensis</i>)	For culms: ☞ Cut culms above one node from the ground level ☞ Do not cut culms less than one year old ☞ Retain 50% of healthy culms in a clump ☞ Remove matured culms from the centre of the clump. ☞ Do not harvest the culms growing on the periphery of the clump. For shoots: ☞ Collect bamboo shoots to a maximum of 25% from a clump ☞ Cut bamboo shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones ☞ After cutting and collecting the bamboo shoot, cover that void place with topsoil ☞ Protect the remaining shoots and let them further develop into culms

5. Map of the NWFP harvesting areas

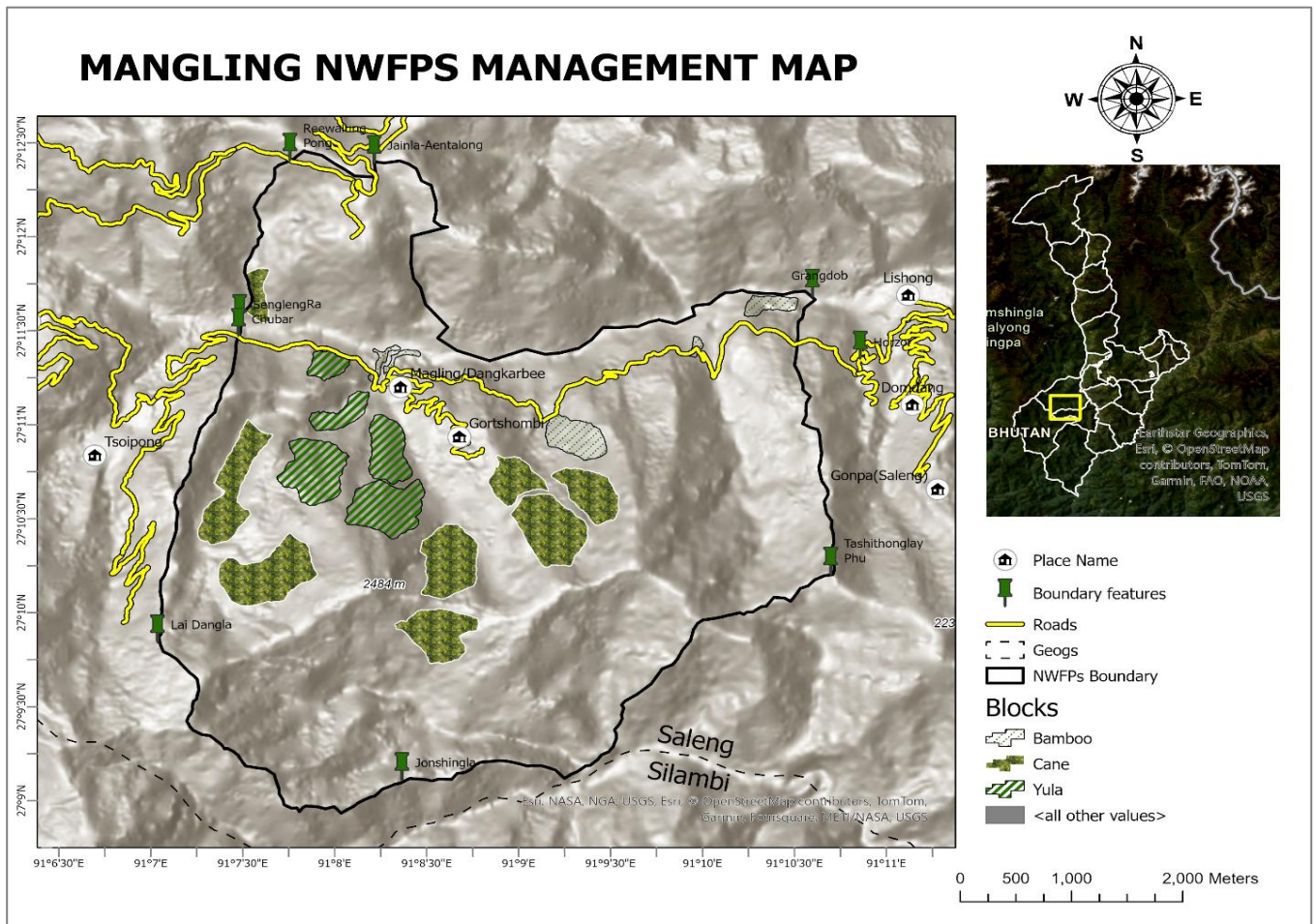


Figure 1: NWFPs harvesting and management map of Mangling

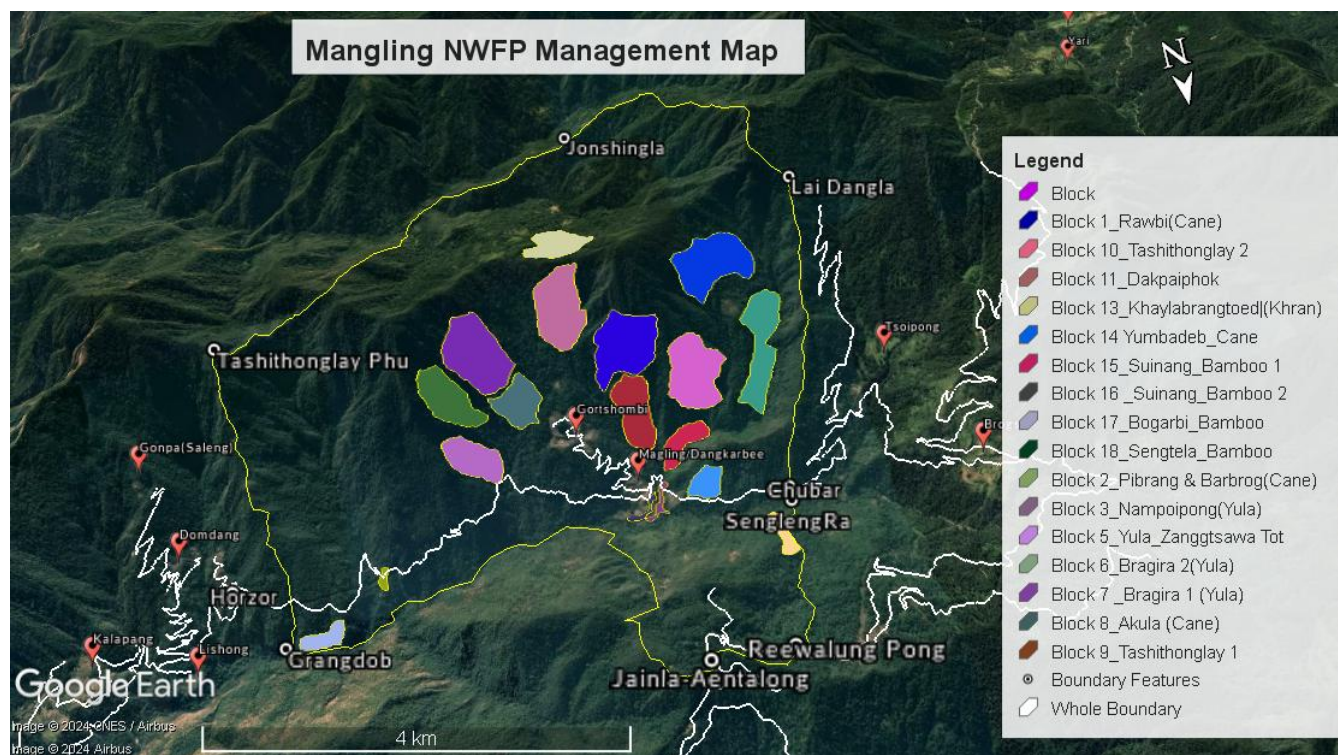


Figure 2: Google Map of the NWFP Harvesting Area of Mangling

Marketing and sales plan

Standard Sizes of Canes	<i>Plectocomia himalayana</i> (Ray), <i>Calamus acanthospathus</i> (Krtah), and <i>Khran</i>: For commercial handicrafts, one headload/backload should consist of 15 meters of cane with varying diameters. - This will be sold at a rate of Nu. 250, which includes the royalty and permit fee.
Yula	- A bundle consists of 4 bundles with a circumference of one meter in solid form, with varying diameters but a length of one internode. - Each bundle will be sold at Nu. 200, including the royalty and permit fee (4 bundles = 1 BL).
Bamboo	- Large bamboo species will be sold at a rate of Nu. 200 per piece, including the royalty. - Small bamboo species (Rewa, Glim) will be sold at a rate of Nu. 50 per piece, including the royalty.
Rubia	Dried Rubia will be sold at a rate of Nu. 70 per kg. Fresh Rubia will not be sold.
Other NWFPs	- NWFPs listed in the annexure will be sold according to the floating rate set by DoFPS. - All NWFPs will be sold only to holders of a trade license. - Commercial extraction of NWFPs is restricted to group members only; outsiders are strictly prohibited from harvesting for commercial or domestic purposes. Harvesting must be approved by the concerned office every quarter.

	ལྷན་སྐྱོང་། The other NWFPs listed in the annexure shall be sold as per the commercial rate fixed by the DoFPS. All the NWFPs shall be sold to the licence holder only.
Permit fee	ལྷན་སྐྱོང་། Permit fee and royalty of NWFPs should be paid as per the royalty quantity of the DoFPS.
Standard 1 BL of cane stem=15m length variable sizes 1 BL of cane shoot= 15 numbers=15m length variable size 1 BL= 4 bundles (1 bundle= 1m circumference)	

The marketing and sales plan for different types of NWFPs that are to be collected by the group, its potential markets where the group shall sell products with approximate quantity, price per unit and the total value of the expected sales are described in the below table:

Table 3: Marketing and sales plan

NWFP.	Potential market	Estimated quantity for sale per season	Unit	Expected price per	Total (Nu)	Remarks
<i>Calamus aconthospathus</i> (Krath/Minji); <i>Calamus latifolius</i> (Drey/Khran); <i>Plectocomia himalayana</i> (Rey/Phashi)	Within Bhutan	700	BL/HL	250	175000	
Cane shoot (<i>Plectocomia himalayana</i> (Rey/Phashi)	Within Bhutan	1000	Nos.	50	50000	
Yula/Ringshoo (<i>Neomicrocalamus andropogonifolius</i>)	India	2500	BL/HL	800	2000000	
Large Bamboo sp	Within Bhutan	300	Nos.	200	60000	
Rawa (<i>Chimonobambusa callosa</i>)	Within Bhutan	1000	Nos.	50	50000	
Rhue (<i>Borinda grossa</i>)	Within Bhutan	100	Nos.	50	5000	
Hekchung (Small bamboo)	Within Bhutan	200	Nos.	50	10000	
Dew yangka (Small bamboo)	Within Bhutan	50	Nos.	50	2500	
Glim (<i>Cepalstachyum</i>)	Within Bhutan	500	Nos.	70	35000	
Dai	Within Bhutan	50	Nos.	50	2500	
<i>Rubia cordifolia</i> (tsod)	Within Bhutan	100	Kgs	70	7000	
Damroo (<i>Elatosteme lineolatum</i>)	Within Bhutan	50	Bundle	20	1000	
Zeelema/Nakey (<i>Diplazium esculentum</i>)	Within Bhutan	100	Bundle	20	2000	

Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)-fresh bark	Within Bhutan	100	kg	250	25000	
Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)-dried bark	Within Bhutan	50	kg	600	30000	
Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)-processed	Within Bhutan	30	kg	2500	75000	
Piper (Pan)	Within Bhutan	50	Bundle	20	1000	
Thormangla (<i>Tupista nutans</i>)	Within Bhutan	20	Bundle	30	600	
Mushroom (Muu)-fresh	Within Bhutan	150	kg	150	22500	
Mushroom (Muu)-dried	Within Bhutan	100	pkt	100	10000	
Soft Broom (Berumgla)	Within Bhutan	150	Bundle	50	7500	
Hard broom (<i>Trachycarpus fortune</i>)	Within Bhutan	80	Bundle	50	4000	
Sisiraba (<i>Osyris lanceolata</i>)	Within Bhutan	50	pkt	60	3000	
Large Bamboo shoot-fresh	Within Bhutan	60	Nos.	50	3000	
Large Bamboo shoot-Dried	Within Bhutan	50	kg	150	7500	
Chawa (<i>Zyanthozum</i>)	Within Bhutan	50	pkt	50	2500	
Walnut (Nut) (<i>Juglans regia</i>)	Within Bhutan	100	kg	50	5000	
Pilpa (<i>Piper longum</i> / <i>P. pedicelatum</i>)	Within Bhutan	50	kg	150	7500	
Nagglabma/Yongjiba (<i>Pouzolzia sanguinea</i>)	Within Bhutan	200	kg	70	14000	

Note: The marketing and sales plans for NWFPs list in the above table (qty) is estimated for 5 years. Only the NWFPs collected extensively are included for marketing and sales plan but the group shall collect other NWFPs list in the annexure as per the demand.

6. Marketing arrangements

The Chairperson, with the support of relevant Forestry Officials, shall gather the demand for NWFPs before the start of any collection season. Subsequently, the marketing of NWFPs will be organized and arranged according to the procedures outlined in Table 4.

Table 4: Procedures for NWFP harvesting, marketing and sharing of income

Activity	A brief description of how the activities are carried out
Harvesting of products	 NWFPs shall be harvested and marketed as a group. Collection should occur simultaneously on the same date and time, as directed by the

	Chairperson. All members should participate at the same collection site for NWFP harvesting.
<i>Post-harvest practices</i>	☞ After harvesting, NWFPs should be cleaned, sorted, and processed according to the standards set by the group to ensure quality and market readiness. ☞ Proper storage facilities should be used to maintain the quality of NWFPs until they are marketed. This includes protecting the products from pests, moisture, and other deteriorating factors. ☞ Packaging should be done in a manner that preserves the quality and facilitates easy transport and handling
<i>Marketing of products</i>	☞ All marketing activities will be coordinated by the Chairperson, ensuring that members adhere to the agreed-upon schedule and procedures. ☞ Members will deliver their products to the designated location on the date agreed upon with the buyer. ☞ Whenever feasible, an open auction will be organized to ensure transparent pricing and fair competition among buyers. ☞ All transactions will be documented, including buyer details, quantities sold, and prices agreed upon, to maintain transparency and accountability. ☞ Regular feedback from buyers will be sought to improve product quality and address any market demands or concerns
<i>Identification of buyers</i>	☞ Buyers must hold a valid license for purchasing NWFPs. ☞ Preference should be given to buyers with a strong reputation for fair trade practices and timely payments. ☞ Buyers should have the capacity to purchase large volumes, ensuring a stable and continuous market for the products. ☞ Buyers should demonstrate a commitment to sustainable and ethical sourcing practices.
<i>Price fixing</i>	☞ Buyers should offer competitive prices that reflect the market value of the NWFPs. ☞ Buyers must agree to the payment terms set by the group, including no advance payments and full payment upon delivery ☞ Members are not allowed to accept advance payments from buyers. However, they have the right to sell to any licensed buyer who offers a better price to the group
<i>Income sharing</i>	☞ 60% of the income from the sale of the NWFPs shall be shared among the group members no matter how much quantity of the products individual members have collected and 40 % will be deposited in the group bank account.
<i>Raising of community development fund</i>	☞ The 40 % of the total income generated shall be deposited in the group savings account
<i>Payment to committee members for their services</i>	☞ Executive members shall receive an incentive of Nu. 150/- per occurrence for each sale of NWFPs conducted within the village (<i>refer to By-laws, section 13</i>).

7. Benefits to the members

- ☞ All group members will have equal rights to collect Non-Wood Forest Products (NWFPs).
- ☞ 20% of the income from product sales, including royalty, shall be deposited into the group account on a quarterly basis.

Table 5: Financial projection and benefits to the members from NWFP sales

Sl.#	Particulars	Product (names)					Total amount
		Cane (BL)	Cane shoot (Nos.)	Yula (BL)	Rubia (kg)	Pipla (kg)	
1	Revenues						
a	Total quantity (unit)	700.00	1,000.00	2,500.00	100.00	50.00	
b	Market price (Nu per unit)	250.00	50.00	50.00	70.00	150.00	
c	Total revenues from forest product sales (a*b)	175,000.00	50,000.00	125,000.00	7,000.00	7,500.00	364,500.00
A	Total revenues	175,000.00	50,000.00	125,000.00	7,000.00	7,500.00	357,000.00
2	Expenses						
a	Packaging materials	0	0		0		0
b	Royalty	2,800.00	4,000.00	2,500.00	1,600.00	1,000.00	11900
c	Transportation costs	0	0	0	0	0	0
d	Compensation fee to executive committee for marketing products	0	0	0	0	0	0
e	Contribution to community development fund	35,000.00	10,000.00	25,000.00	1,400.00	1,500.00	71,400.00
f	Incidental expenses	0	0	0	0	0	0
B	Total expenses	37800	14000	27500	3000	2500	440,300.00
C	Total benefit (net-income) of group members (total revenues A – total expenses B)	137,200.00	36,000.00	97,500.00	4,000.00	5,000.00	279,700.00
D	Average income to group members (C/ no of group members)	5,716.67	1,500.00	4,062.50	166.67	208.33	4,070.06
E	Price per unit paid to group members (C/1a)	196.00	36.00	39.00	40.00	100.00	

8. Benefits to the group

- ☞ All group members shall have equal rights to collect and market Non-Wood Forest Products (NWFPs).

- ☉ The group shall benefit from membership fees and a 20% deduction from the sale value of raw NWFPs collected and sold by individual members.
- ☉ The budget shall be allocated for use during local Tshechus and for the development of the local lhakhang.
- ☉ For finished products, a 20% deduction from the sale value shall be deposited from individual marketing efforts.
- ☉ Develop a sense of ownership over the resources and provide their support and commitments in the conservation, protection, and sustainability of resources.

Table 6: Financial projection of the benefits to the group, the community development fund

Sl. #	Particulars	Amount (Nu)
	Revenues	
1	Membership fees	21,600.00
2	Contribution to the Community Development Fund (CDF) from sales of product 1 (Cane)	175,000.00
3	Contribution to the Community Development Fund (CDF) from sales of product 2 (Cane shoot)	50,000.00
4	Contribution to the Community Development Fund (CDF) from sales of product 3 (Yula)	125,000.00
5	Contribution to the Community Development Fund (CDF) from sales of product 4 (Rubia)	7,000.00
6	Contribution to the Community Development Fund (CDF) from sales of product 5 (Pipla)	7,500.00
7	Fines for absentees in group meeting	-
A	Total revenues	386,100.00
	Expenses (only include expenses that will be paid from the group funds)	
1	Stationeries	1,500.00
2	Working lunch & Refreshment during meeting and visitors	2,000.00
3	Transportation (Lumpsum)	5,000.00
B	Total Expenses	8,500.00
C	Balance: Total revenues – total expenses	377,600.00

9. Resources Assessment

Resource assessments for cane species (*Plectocomia himalayana*, *Calamus acanthospathus*, and *Calamus latifolius*) were conducted across 214.10 hectares in 9 blocks, while yula (*Neomicrocalamus andropogonifolius*) was assessed over 92.95 hectares in 5 blocks. Details of the area and sampling plots are provided in Table 7.

Circular plots with a radius of 12.62 meters were used for the sampling. It is recommended to inventory at least 30% of the total population size (N). If funding and time permit, conducting the maximum number of sample plots is advisable to obtain more precise results. A stratified systematic grid sampling method was

employed for this assessment, with grids generated using GIS Software (QGIS). The grid distance was set at 150 meters by 150 meters.

Table 7: Overview of the cane and bamboo blocks area

Block	Block Name	Produces	Area (Ha)	Sample plots
Block 1	Rawbi	Cane	36.59	19
Block 2	Pibrang & Barbrog	Cane	31.38	21
Block 3	Nampoipong	Yula	28.07	12
Block 4	Nampoipongjug	Yula	9.96	20
Block 5	Zangitsawatot	Yula	7.44	19
Block 6	Bragira 1	Yula	19.04	8
Block 7	Bragira 2	Yula	28.44	13
Block 8	Akula	Cane	29.10	18
Block 9	Tashithonglay 1	Cane	32.82	20
Block 10	Tashithonglay 2	Cane	19.79	20
Block 11	Dakpaiphok	Cane	16.72	7
Block 12	Kradari	Cane	15.28	18
Block 13	Khaylabrangtoed	Cane	7.64	20
Block 14	Yumbadeb	Cane	24.78	24
Block 15	Suinang	Bamboo	1.85	17
Block 16	Suinang	Bamboo	2.62	21
Block 17	Bogarbi	Bamboo	6.67	Not Assessed
Block 18	Sengtela	Bamboo	1.50	Not Assessed

10. Analysis of Resource Assessment Data

11.1. Cane resources

Table 8 shows the result of over annual harvesting limits for cane harvesting over 10 years in the Mangling NWFP area. The block-wise AHL analysis of cane is given in the Annexure.

Table 8: Overall AHL for cane harvesting

Annual harvesting limits of canes in number of stems and backloads over a time period of 10 years for all the cane blocks					
Time period for harvesting canes	Cane Species	Sub-block area (Ha)= (a)		214.10	
		Sample Plots (N)		160	
		Sampling area covered (Ha) = (b)		8.00	
		Number of canes in sampling area (c)	Canes per hectare/ acre (d) (= c / b)	AHL of cane stems per sub-block (e) (= a *d)	AHL of back loads per sub-block (f) (= e / average number of canes per backload)
Year 1-2: Present estimated length: ≥ 15m	<i>Calamus acanthospathus</i> (Krath)	102	13	2730	455
	<i>Calamus latifolius</i> (Gray)	7	1	187	31
	<i>Plectocomia himalayana</i> (Patsha)	920	115	24622	4104
Year 3-4: Present estimated length:10-14.9m	<i>Calamus acanthospathus</i> (Krath)	55	7	1472	147
	<i>Calamus latifolius</i> (Gray)	1	0	27	3
	<i>Plectocomia himalayana</i> (Patsha)	401	50	10732	1073

Year 5-6: Present estimated length 6 -9.9m	<i>Calamus acanthospathus</i> (Krath)	24	3	642	40
	<i>Calamus latifolius</i> (Gray)	22	3	589	37
	<i>Plectocomia himalayana</i> (Patsha)	271	34	7253	453
Year 7-8: Present estimated length 3 - 5.9m	<i>Calamus acanthospathus</i> (Krath)	24	3	642	27
	<i>Calamus latifolius</i> (Gray)	39	5	1044	43
	<i>Plectocomia himalayana</i> (Patsha)	282	35	7547	314
Year 9-10: Present estimated length 1-2.9m	<i>Calamus acanthospathus</i> (Krath)	243	30	6503	163
	<i>Calamus latifolius</i> (Gray)	40	5	1071	27
	<i>Plectocomia himalayana</i> (Patsha)	647	81	17315	433
Total	<i>Calamus acanthospathus</i> (Krath)	448			832
	<i>Calamus latifolius</i> (Gray)	109			141
	<i>Plectocomia himalayana</i> (Patsha)	2521			6377

11.2. Yula (small bamboo) resources

Table 9 shows the result of over annual harvesting limits for yula (small bamboo) harvesting over 10 years in the Mangling NWFP area. The group decided to harvest the all the yula resources during management period because the yula has started to flower reaching for the maturity.

Table 9: Overall Yula harvesting Limit

Sub-Block		Sample plots		No of Yula clums to be harvested this year			No. of Yula clums harvested last year			Average no. of culms per backload=[i]	Range of AHL of Yula backloads[j]=(e/i-h/i)]
Name	Area (Ha)=[a]	No.	Total area (Ha)=[b]	in plots=[c]	per Ha [d=(c/b)]	in sub-blocks [e=(d*a)]	in plots=[f]	Per Ha [g=(f/b)]	in sub-blocks [h=(g*a)]		
B3	28.07	12	0.62	32	51	1440	6	10	270	36	33
B4	9.96	20	1.00	505	505	5032	0	0	0	7	686
B5	7.44	19	0.95	381	401	2983	30	32	235	12	226
B6	19.04	8	0.42	45	106	2025	0	0	0	38	53
B7	28.44	13	0.63	384	608	17280	47	74	2115	10	1559
Total											2556

11.2. Large bamboo resources

The wild large bamboos were assessed for two blocks and the annual harvesting limit was calculated as detailed in the Table 10. The homestead bamboos were also assessed in the Mangling village and the detailed are furnished in the Table 11.

Table 10: AHL for wild large bamboos

Block	Block area (Ha)	Area Coverage (m ²)	No. of culms to be harvested (AHL)			
			First Year (≥3 years)	Second Year (≥2 years)	Third Year (≥1yr)	Shoot (<1yr)
B15	1.85	247.94	64	63	48	113
B16	2.62	116.85	34	15	18	118
Total			98	78	66	231

Table 11: AHL for homestead bamboos of Mangling village

Species	Area (m ²)	Year 1	Year 2	Year 3	Year 4
		No. of Culm ≥3 years	No. of Culm ≥2 years	No. of Culm ≥1yr	No. of Culm <1yr (Shoot)
<i>Bambusa hookeriem</i>	0.08	0	12	6	9
<i>Bambusa nuttans</i>	324.15	159	178	152	179
<i>Dendrocalamus hamiltonii</i> (Sue)	214.77	51	83	80	104
<i>Dendrocalamus sikkimensis</i>	2.86	53	69	42	24
<i>Dendrocalamus strictus</i>	1.19	5	11	16	13
Grand Total	543.06	268	353	296	329

PART B: GROUP'S BY-LAWS

12. NWFP management and marketing group	
12.1 Membership arrangement and resignation	Membership Arrangement ☞ All residents and fragmented households/families of Mangling Village shall be eligible for membership upon payment of the cumulative annual membership fee of Nu. 1200 per year. ☞ Individuals who choose not to become members are not allowed to use these resources for commercial purposes. However, they may use these resources for personal, non-commercial purposes after consulting with the Executive Members. ☞ If any household is found using Non-Wood Forest Products (NWFPs) for commercial purposes, such cases will be handled according to the existing rules and regulations. Membership resignation ☞ Members who choose to withdraw may do so at their discretion. By withdrawing voluntarily, they will lose any right to Non-Wood Forest Products (NWFPs). Additionally, any fees paid or labour contributions made during their membership will not be refunded. They will also not be entitled to receive an equal share of the group's total accumulated funds. ☞ Disadvantaged members who are required to leave the group involuntarily, such as due to having only one family member, may do so. However, in such cases, they will not be entitled to receive an equal share of the group's total accumulated funds.
12.2. Responsibilities and rights of members:	☞ All members must responsibly manage Non-Wood Forest Products (NWFPs). ☞ NWFP harvesting should only occur during the appropriate seasons specified in the management plan. ☞ Members must follow and respect the instructions of the executive members. ☞ Members must watch for and report any illegal NWFP harvesting by non-members to the forest office or executive members. ☞ All members have the right to harvest and sell NWFPs as per this plan.

13. Executive committee of NWFP management and marketing group

13.1 Chairperson	Name of Chairperson: Rinchen Pelmo (17597037) Responsibilities of the Chairperson:  The Chairperson holds the authority to impose fines and penalties on members who do not comply with his/her directives regarding meetings and timing.  The Chairperson is responsible for obtaining the NWFPs collection permit from the forest office during the collection season.  The Chairperson must inform or announce the date of NWFPs collection to all members.  The Chairperson shall coordinate the harvesting of NWFPs.  The Chairperson shall contact buyers/traders to determine the required quality of NWFPs and inform the members accordingly.  The Chairperson is responsible for seeking out buyers/traders for the NWFPs.  The Chairperson is to lead and coordinate all meetings related to this plan.  The Chairperson is responsible for resolving disputes and conflicts among members.  The Chairperson must manage overall financial arrangements.  The Chairperson is required to attend or participate in trainings, workshops, or study tours as necessary.
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13.2. Secretary	Name of Secretary: Chador Mo (17900638) Responsibilities of Treasurer:  Ensure comprehensive documentation of all cash transactions.  Collect membership fees, incentives, and compensation fees, and ensure their prompt deposit in the bank.  Supervise all financial transactions in collaboration with the Chairperson and Secretary.  Assume the responsibilities of the Secretary when necessary.  Maintain meticulous records of sales data related to Non-Wood Forest Products (NWFP).  Present a detailed account statement to the group during meetings.  Issue receipts for all collected fines and penalties.
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11.3 Treasurer	Name of Treasurer: Rinchen Zangmo (17318338) Responsibilities of Secretary:  Maintain comprehensive records of all sales transactions.  Document the minutes of all meetings accurately and in detail.  Participate in and contribute to trainings, workshops, and study tours as mandated.  Assist the Chairperson in fulfilling their duties.
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14. Tenure of committee member

-  The appointed committee shall serve the group for a term of five years. Should an executive member fail to fulfil their responsibilities or violate the by-laws as outlined in this plan, the group

shall have the authority to re-select the executive members.

15. Incentives to the committee members

- 15. The group members have decided that no compensation or incentives shall be paid to the executive members. However, if sufficient funds are available, any incentive shall be determined by agreement among the group members during a general meeting.

16. Sources of generation of group fund

16.1 Membership registration fees.	16. Each member is obligated to contribute an annual membership fee of Nu.1200/- , payable during the annual meeting, for the entire duration of the plan.
16.2 Collection from sale of NWFP produces/products	16. According to the Group By-Laws, an amount equivalent to 20% of the gross sales, inclusive of royalty fees, shall be deposited into the group account quarterly.
16.3. Fines and Penalties	16. Fines and penalties shall be imposed for violations of the By-laws and non-compliance with the management plan as outlined in the following section.
16.4. Other (not listed)	16. The funds will be accrued from sources not listed above, including donations and contributions from visitors and donors.

17. Group fund management and utilization

17.1. Administrative cost	17. A portion of the funds shall be allocated for the management of NWFP resources, including activities such as plantation and nursery development.
17.2. NWFP resource development	
17.3. Welfare to members during natural calamities.	17. The group shall allocate 20% of the revenue from the sale of NWFPs to community development funds, which shall be deposited into the group's account.
17.4. Welfare to disadvantaged groups in the group/in the village.	17. The group shall offer loans to its members at an interest rate of 6% per annum, with a loan term of one year. A formal loan agreement shall be executed between the borrower and the group.
17.5. Community cultural activities	17. These funds shall be utilized exclusively for community development activities.

18. Conflict Resolution

- 18. In the event of any disputes or disagreements arising within the NWFP Management Group which cannot be amicably resolved by its members internally, such matters shall be escalated to the Chairperson for prompt resolution.
- 18. Should the issue persist unresolved at the group level, it shall subsequently be referred to the Gewog Administration through the Chiwog Tshogpa.
- 18. In cases involving conflicts between the NWFP Group and external parties, the Chairperson shall forward the matter to both the Divisional Forest Office and the Gewog Administration for necessary action.

19. Meeting and purpose

- ✓ The meetings shall held quarterly (4 times in year) to address matters concerning progress evaluation, strategic planning, and seeking approval of the collection marketing of NWFPs.

20. Penalties

- ✓ Any member who sells NWFPs directly to traders without notifying the members of the Group Executive Committee shall be subject to penalties per the FNCRR, 2023.
- ✓ Non-members found engaging in illegal acquisition or trade of NWFPs shall be handled per the FNCRR, 2023.
- ✓ Group members must sell NWFPs only to valid license holders. Failure to comply shall result in penalties as stipulated in the FNCRR, 2023.
- ✓ Members who are absent from meetings shall be required to pay a fine of Nu.500/-.
- ✓ Committee members who fail to attend meetings shall be fined Nu.800/-. The collected amount shall be deposited into the group savings account, with exceptions made for circumstances of death or severe illness.
- ✓ If any executive member misappropriates group funds, they are liable to repay double the misused amount within 15 days.
- ✓ A late fee of Nu. 50/- shall be imposed for every 10 minutes of delay in group-related work or meetings, accruing cumulatively after the initial 10 minutes.
- ✓ Offences not specified shall be addressed per the FNCRR, 2023.
- ✓ Offenses related to NWFP management not explicitly listed above shall be handled according to existing rules and regulations pertaining thereto.

21. Powers of the Department

- ✓ The Department reserves the authority to terminate the NWFP management and marketing group in the event of any breach of the management plan's provisions or an inability to execute activities per said plan. According to Section 145 of FNCA, 2023, any NWFP Management Group that fails to adhere to the NWFP Management Plan may be subject to suspension, reinstatement, or termination per procedures stipulated in the Rules.

22. Cancellation of NWFP Management Certificate

Under Chapter III, sections 156, 157 & 158 of FNCRR, 2023:

- ✓ The NWFP Management Certificate shall be revoked if the performance of NWFMG does not conform to the provisions of the approved management plan or if the sustainability of the resource is jeopardized.
- ✓ Should the members of the NWFMG elect to revoke the NWFP Management Certificate according to these Rules, such revocation shall take effect on the date the decision is communicated in writing to the CFO. The NWFMG must then promptly surrender the Certificate to the Department.
- ✓ 3. The CFO retains the authority to investigate the grounds for such revocation to ensure compliance with regulatory requirements and shall notify the NWFMG upon the cancellation of the certificate.

Annexure I: NWFP Group Membership List

Sl. No	Name	CID #	Gung No.	Thram No.
1	Sigyal	10711000259	Ja-11-235	205
2	Sangay	10711000274	Ja-11-236	245
3	Sangay Wangchuk	10711000284	Ja-11-237	360
4	ChadorMo	10711000297	Ja-11-238	493
5	Pema Wangmo	10711000305	Ja-11-239	48
6	Dorji Tshewang	10711000306	Ja-11-239	1010
7	Dorji Wangmo	10711003326	Ja-11-307	205
8	Choten Wangmo	10711000293	Ja-11-342	689
9	Kencho	10711900264	Ja-11-343	684
10	Pema Dema	10711000304	Ja-11-344	681
11	Kezang	10711000275	Ja-11-363	685
12	Rinchen Pelmo	10711000263	Ja-11-390	934
13	Rinchen Zangmo	10711000276	Ja-11-419	1189
14	Ausin	10711000277	Ja-11-420	1190/SAL-3295
15	Tenzin Wangchuk	10711000294	Ja-11-208	218

Annexure II: Block description form for canes

Date: December 2023

Village: Mangling

Name of cane area: Reflected in Table 7

Block name: Reflected in Table 7

Total area (ha): Reflected in Table 7

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle < 45 %
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Forest description	Dominant and associate species, canopy cover: <i>Symplocos sp.</i> , <i>Toona sp.</i> , <i>Quercus sp.</i> , <i>Michelia sp.</i> , <i>Lithocarpus sp.</i> , <i>Perse sp.</i> ,
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Sub-block	No.	Density	Area (ha)	No of sample plots
	1	<i>Rawbi</i>	36.59	19
	2	<i>Pibrang-Barbrog</i>	31.38	21
	3	<i>Akula</i>	29.04	18
	4	<i>Tashithonglay 1</i>	32.10	20
	5	<i>Tashithonglay 2</i>	19.79	20
	6	<i>Dakpaiphok</i>	16.72	7
	7	<i>Kradari</i>	15.28	18
	8	<i>Khaylabrangtoed</i>	7.64	20

	Rawbi		√	
	Pibrang-Barbrog		√	
	Akula	√		
	Tashithonglay 1	√		
	Tashithonglay 2	√		
	Dakpaiphok	√		
	Kradari		√	
	Khaylabrangtoed		√	

	Current management practices in entire block: <i>The first management plan was completed and the NWFPs were managed as per the prescribed management plans.</i>
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	Fire	<i>Since the large portion of management area falls in the cool broadleaf there is less chances of forest fire. However, the two bamboo blocks fall in the mixed chirpine and broadleaf forests</i>
	Grazing	<i>There is grazing pressure in the management areas</i>
	Illegal exploitation	<i>There is illegal exploitation of NWFPs by other outsiders like NRDCL logging labours since the NWFPs management area falls in the production circle.</i>
	Over-exploitation	<i>During the resource assessment, there is no sign of over-exploitation of NWFPs</i>
	Other:	

	Future opportunities for the entire block: <i>The cane, bamboo and other NWFPs can be harvested for the livelihood improvement of the locality. The proper management would regrowth of the NWFPs.</i>
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Annexure III: Block description form for Yula

Date: December 2023

Village: Mangling

Name of cane yula: Reflected in Table 7

Block name: Reflected in Table 7

Total area (ha): Reflected in Table 7

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle < 45 %
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Forest description	Dominant and associate species, canopy cover: <i>Symplocos sp., Toona sp., Quercus sp., Michelia sp., Lithocarpus sp., Persea sp.,</i>
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Sub-block	No.	Density	Area (ha)	No of sample plots
	Reflected in Table 7			

Plantation/ enrichment potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	Nampoipong	√		
	Nampoipongjug	√		
	Zangitsawatot			√
	Bragira 1	√		
	Bragira 2	√		

Current management	Current management practices in entire block: <i>The first management plan was completed and the NWFPs were managed as per the prescribed management plans.</i>
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Current threats	Threat	Description
	Fire	<i>Since the large portion of management area falls in the cool broadleaf there is less chances of forest fire. However, the two bamboo blocks fall in the mixed chirpine and broadleaf forests</i>
	Grazing	<i>There is grazing pressure in the management areas</i>
	Illegal exploitation	<i>There is illegal exploitation of NWFPs by other outsiders like NRDCL logging labours since the NWFPs management area falls in the production circle.</i>
	Over-exploitation	<i>During the resource assessment, there is no sign of over-exploitation of NWFPs</i>
	Other:	

Opportunities	Future opportunities for the entire block: <i>The cane, bamboo and other NWFPs can be harvested for the livelihood improvement of the locality. The proper management would regrowth of the NWFPs.</i>
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Annexure IV: Data Compilation Sheet for canes

Block	Calamus acanthospathus (Krath)					Calamus latifolius (Gray)					Plectocomia himalayana (Patsha)				
	> 15 m	10-14.9m	6 -9.9m	3 - 5.9m	1-2.9m	> 15 m	10-14.9m	6 -9.9m	3 - 5.9m	1-2.9m	> 15 m	10-14.9m	6 -9.9m	3 - 5.9m	1-2.9m
B1	0	0	0	0	9	0	0	0	0	0	123	69	55	58	80
B2	41	15	6	13	41	0	0	0	0	19	37	0	23	0	19
B5	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
B8	0	0	0	0	0	0	0	0	0	0	99	41	18	36	133
B9	2	0	0	1	0	0	0	0	0	0	239	71	43	96	127
B10	46	26	7	7	132	0	0	0	0	0	79	50	35	32	91
B12	13	14	11	3	61	0	0	0	0	0	12	19	3	8	40
B13	0	0	0	0	0	7	1	22	39	39	0	0	0	0	0
B15	0	0	0	0	0	0	0	0	0	0	331	151	94	52	157
Total	102	55	24	24	243	7	1	22	40	58	920	401	271	282	647

Annexure V: Data compilation sheet for Yula

Blocks	Total No of Culms	Culms to be harvested this year	Culms harvested last year	Estimated no. of culms in 1 back-load
B3	73	32	6	36
B4	893	505	0	7
B5	720	381	30	12
B6	435	45	0	38
B7	579	384	47	10